FEE 447 (COMPUTER HARDWARE MAINTENANCE AND REPAIR) LECTURE NOTE

This course is about computer hardware repair and maintenance. It is intended to analyze and explain the hardware components part of the computer. It is also to serve as a guide to the assembly ~~and~~ installation, ~~and~~ diagnosing and troubleshooting computer systems. This course will also give an insight to both hardware and hardware safety preventive maintenance procedure.

1.0 COMPUTER HARDWARE COMPONENTS

COMPUTER HARDWARE is the equipments involved in the function of a component and consist of components that can be physical handled. The functions of these componens are typically divided into three main categories:

INPUT, OUTPUT AND STORAGE. Component in these categories connect to microprocessors, specifically, the computers center processing unit (CPU), the electronics circuitry that provides the computational ability and control of the computer, via wires or circuitry called *BUS*.
Software on the other hand is the set of instruction a computer uses to manipulate data, such as word processing program or a video game. These programs are usually stored and transferred via the computer hardware to and from a storage device. The interaction between input and output hardware is controlled by software called basic input output system software (BIOS).
Although microprocessors are still technically considered to be hardware, portion of their function are also associated with computer software since microprocessor have both hardware and software aspects they are therefore often referred to as *FIREWARE.*

1.1 INPUT DEVICE

Input device permits the computer to communicate with the computer. Input hardware consist of external devices that is, components outside of the computer CPU that provides information and instruction to the computer. Some of the devises includes:

KEYBOARD: A typewriter like device that allows the user to type in text and commands to the computer. Some keyboards have special functions keys or integrated pointing devices, such as trackball or touch sensitive region that let the user's finger motions move an on screen cursor.

MOUSE: A pointing device designed to be gripped by one hand. it has a detector device (usually a ball) on the bottom that enabled the user to control the motion of an on screen pointer, or cursor by moving the mouse in a flat surface. As the device moves across the surface, the cursor moves across the screen. To select item or choose commands on the screen, the user presses a button in the mouse.

JOYSTICK: A pointing device composed of a lever that moves in multiple directions to navigate a cursor or other graphical object on a computer screen.

LIGHT PEN: A stylus with a light sensitive tip that is used to draw directly on a computer video screen or to select information on the screen by pressing a clip in the light pe or by pressing the light pen against the surface of the screen. The pen contain light sensor that identify which portion of the screen it is passed over.

OPTICAL SCANNER: This uses light sensing equipment to convert image such as pictures or text into electronics signals that can be manipulated by a computer. For example, a photograph can be scanned into a computer and then included in a text document created on that computer. The two most common scanner type are *flatbed scanner*, which is similar to an office photocopier and the *handheld scanner* which is passed normally across the image to be processed.

DIGITIZER: A flat plastic rectangle with subsurface electronics, used in conjunction with a pointing device in many engineering and design applications as well as illustration work. When the pointing device is moved on the surface of the tablet, the location of the device is translated to a specific on screen cursor position.

MICROPHONE: A device for converting sound into signals that can then be stored, manipulated, and played back by the computer.

VOICE RECOGNITION MODULE: A device that converts spoken words into information that the computer can recognize and process.

1.2 OUTPUT DEVICE:

Output devices are the equipment that the computer uses to make its results available to the user. Output hardware consist of external devices that transfer information from the computer CPU to the computer user. Some of these devices are:

MONITOR (VDU): A video display, or screen, that converts information generated by the computer into visual information. Displays commonly take one of two forms; a video screen with a cathode ray tube (CRT) or a video screen with a liquid. Crystal display (LCD). A CRT based screen, or monitor, look similar to a television set. Information from the CPU is displayed using a beam of electrons that scans a phosphorescent surface that emits light and create images. An LCD based screen displays visual information on a flatter and small screen than a CRT based video monitor. LCDs are frequently used in laptop computers.

PRINTER: A device that takes text and image from a computer and prints them on paper. Dot matrix printers use tiny wires to impact upon an ink ribbon to form characters. Laser printers employ beam of light to draw images on the drum that then picks up fine black particles called *toner*. The toner is fused to a page to produce an image. Inkjet printers fire droplets of ink onto a page to form characters and pictures

1.3. INPUT/OUTPUT DEVICES

Input/output device are piece of hardware that are used for both providing in formation to the computer and receiving information from it. An input/output device transfer information in one of two directions depending on the current situation. Example of such devices is:

MODEM: A device that connects a computer to a telephone line or cable television network and allows information to be transmitted to or received from another computer. Each computer that sends or receive information must be connected to a modem. The digital signal send from one computer is converted by the modem into analog signal, which is then transmitted by the telephone line or transition cable to the receiver modem which convert signal back into a digital signal that the receiver computer can understand.

DISK DRIVE: A device that reads or writes data, or both on a disk medium. The disk medium may be either magnetic as with floppy disks or hard disks; optical as with CD-ROM (Compact disc – read only memory) disk; or a combination of the two as with magneto- optical disks. A disk drive is both an input and an output device because it can either provide stored information or store the data after processing.

1.4 STORAGE DEVICE:

Storage devices are those in or on which computer information can be kept. Storage hardware provides permanent storage of information and programs for retrieval by the computer. The two main types of storage devices are disk drives and memory. Disk drive read or writes data on a disk medium, while memory refers to the computer chips that store information for quick medium, while memory refers to the computer chips that store information for quick retrieval by the CPU. Some example of storage devices include:

HARD DISK DRIVE: A device that store information in magnetic particle embedded in a disk. Usually a permanent part of the computer, hard disk drive can store large amounts of information and retrieve that information very quickly.

FLOPPY DISK DRIVE: A device that store information in magnetic particles embedded in removable disks that may be floppy or rigid. Floppy disks store less information than a hard disk drive and retrieve the information at much slower rate.

MAGNETIC OPTICAL DISK DRIVE: A drive that store information on removable discs that are sensitive to both laser light and magnetic field. They can typically store as much information as hard disks, but they have slightly slower retrieval speeds.

COMPACT DISC DRIVE: A device that store information on part burned into surface of disc of reflective materials. CD-ROMs can store much information as a hard disc but have a slower rate of information retrieval.

DIGITAL VIDEO DISK (DVD): A device that looks and work like a CD-ROM but can store more than 15 times as much information.

RANDOM ACESS MEMORY (RAM): A set of chips used to store the information and instruction to operate the computers programs. Typically, programs are transferred from storage on a disk drive to RAM. RAM is also known as volatile memory because the information within the computer chips is lost when power to the computer is turned off.

READ ONLY MEMORY (ROM): A set of chips that contain critical information's and software that must be permanently available for computer operation, such as the operating system that directs the computes action from start up to shut down. ROM is called non volatile memory because the memory chips do not lose their information when power to the computer is turn off.

1.5 THE CENTRAL PROCESSING UNIT (CPU)

The CPU may be a signal chip (microprocessor) or a series of chips that perform arithmetic and logical calculations and control operation of the other elements of the system. Most CPU chips and microprocessors are composed of four functional sections: *an arithmetic/logical unit, register, control section and an internal bus.*

ARITHEMATIC/LOGIC UNIT: the part of the CPU that gives the chip its calculatary ability and permits arithmetic and logical operations.

REGISTERS: Temporary storage areas that hold data, keep track of instructions, and hold the location and results of these operations.

CONTROL SECTION: This section has three principal duties. It times and regulates the operations of the entire computer system, its instruction decoder reads the patterns of data in a designated register and translated the pattern into an activity, such as adding and comparing, and its interrupt unit indicates the order in which individual operations use the CPU and regulates the amount of CPU time that each operation may consume.

INTERNAL BUS: A network of communication lines that connect system. The three types of CPU buses are:

- **CONTROL BUS:** This consist of a line that sense input signal and another line that generates control signal from within the CPU.
- **ADDRESS BUS:** This is one way line from the processor that handles the location of dats in memory addreses.
- **DATA BUS:** A two way transfer lines that both reads data from the memory and data into memory.

20

1.6 - HARDWARE CONNECTIONS:

To function, hardware requires physical connections that allow components to communicate and interact. This is usually achieved using cable or circuitry and may be categorized as follows:

- **BUS CONNECTION:** This provides a common interconnected system composed of a group of wire or circuitry that coordinate and moves information between the internal parts of a computer. A bus is characterized by two features; how much information it can manipulate at one time, called the bus with bus width and how quickly it can transfer these data.
- **SERIAL CONNECTION:** A wire or set of wire used to transfer information from the CPU to an external device such as a mouse, keyboard, modem, scanner and some types of printers. This type of connection transfers only one piece of data at a time and is therefore slow. The advantage ~~to~~ of using a serial connection is that it provides effective connections over long distances.
- **PARALLEL CONNECTION:** Multiple sets of wires used to transfer blocks of information simultaneously. Most scanners and printers use this type of connection. A parallel connection is much faster than a serial connection, but it is limited to distance of less than 3m (10ft) between the CPU and external device.

1.7 DEFINATIONS

Here are the definitions of some terms related to computer hardware.

- **PERIPHERAL:** A piece of computer hardware, such as a disk drive, printer modem or joystick, that is external ~~to~~ but controlled by a computers central processing unit although peripheral often implies. Additional but not essential " many peripheral devices are critical elements a fully functioning and useful computer system"
- **ACCESSORY:** A peripheral or add on to a computer, such as a mouse or a modem. An accessory provides a function not available on the original machine, but it is not necessary to the operation of the machine.
- **BIT:** Abbreviation for binary digit, the smallest unit of information stored in a computer or peripheral device. A bit is represented by the number 1 and 0, which correspond to the states on and off, true and false, or yes or no. bits are the building blocks for all information processing that goes on in digital electronics and computers.
- **BYTE:** A group of eight bits of computer information, representing a unit of data such as a number or letter. It also signifies a unit of computer memory equal to that needed to store a single character. Bits and bytes are the basis for creating all meaningful information and programs on computer. Byte is the key unit for measuring quantity of data. Data quantity is commonly measured in kilobyte (1024byte), megabytes (1,048,576byte) or gigabyte (about 1billion bytes).
- **CHARACTER:** A unit of computer data that signifies a single letter, number or symbol that can be display on a computer screen or printer and represent one byte of data. Each sequence of bits in the byte encodes a specific unit of information.

- **WORD:** The fixed or maximum number of bits processed as a single unit by a computer. Most computers operates by manipulating groups of 2,4, or 8.

2.0 INSIDE THE PERSONAL COMPUTER

This section deals mainly with the parts of the computer inside the case. Starting with the computer case and the power supply, mother board, microprocessor, memory, drive controller, video/sound cards and disk drives are discussed. Next, peripherals such as monitors, keyboards and the mouse as well as the different and connectors on the back of the computer and covered.

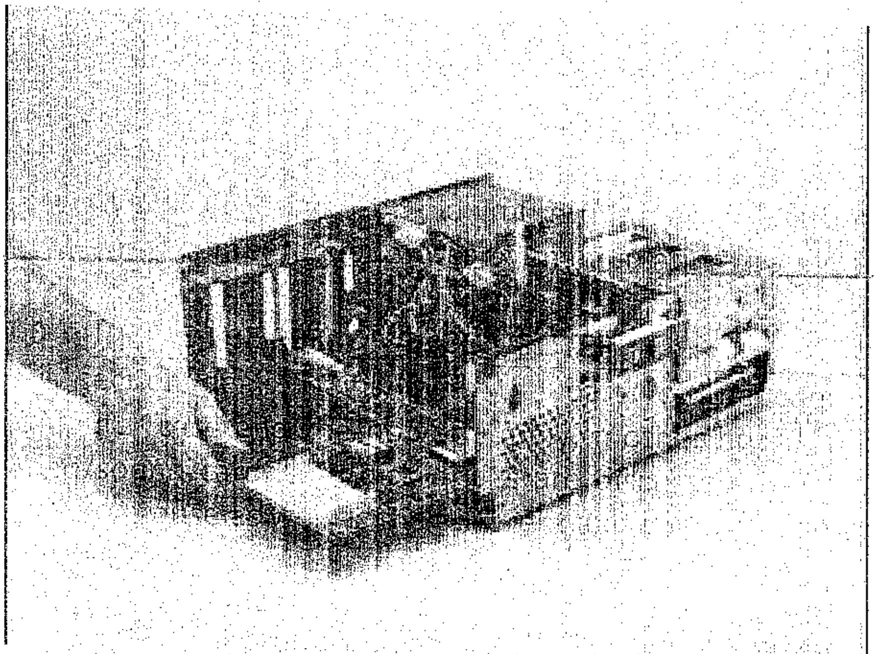


Fig 2.1: INSIDE A PERSONAL COMPUTER

2.1 THE CASE

There is tool basic style of cases the computer; the tower and the desktop style case. *Desktop style* is in the shape of a rectangular box that sets flat on a desk. Usually the computer monitor is placed on top of it. A tower case looks similar to a tower as the name says. There computers will be placed off to the keyboard and monitor. The tower case is the most popular style of desktop computer today mainly because it can be design for better heat dissipation. Tower cases come in several sizes which are:

- **MINI TOWER:** This is the most popular and smallest size of the tower cases (can be less than 14 inches high). It comes with a power supply of 200-250 watts and cools a little better than a desktop case. It has two 3.5" and two 5.25" external drive bays, on or two internal drive bays and is a little cramped inside.
- **MID TOWER:** The standard size (approx 17-20 inches in height), recommended for most application including standard desktop systems and some servers. It comes with a power supply of 200-300 watts and has two 3.5" and three 5.25" external drive bays and two or

three internal drive bays it has less room than a full tower to work inside, cools really well and still have room for expansion.

- **FULL TOWER:** The largest size (up to 36" high) that sits on the floor and is usually used for high powered server. It comes with power supplies of 350 watts or more, may have two 3.5" and four or more 5.25" external drive bays and may have four or more internal drive bays. It is very roomy inside, has better cooling and is the most expensive.

In tower casing, the front panel contains among other things, the external drive bays (CDROM drive, floppy disk drive and spares); power and reset switches as the case may be. The back panel has two power connectors, two ps/2 connectors and slots covered b a metal plate for add on cards like the monitor, modem e.t.c

The exact locations of many of these items vary somewhat from computer to computer, but the overall layout is generally the same. Types of cases come to fit AT and ATX form factor (sizes). The AT or ATX version refers to the type of motherboard the case is designed to fit. The at case is for the old type of motherboards such as for the 80486- Pentium 266 range of microprocessor based computers.

2.2 THE MOTHERBOARD

The motherboard is the large circuit board inside the computer case. It is the mother of all boards in the computer system and is sometimes called the system board, the baseboard or less commonly, the planar board. Everything connected to the computer system, plugs either directly or indirectly into the motherboard.

The motherboard is the central nervous system of the computer system. It contains the CPU, the BIOS ROM chip (Basic input/output system), and the CMOS setup information. It has expansion slots for installing different adapter cards like your video card, sound card, network interface card and modem. This circuit board provides a connector for the keyboard as well as housing the keyboard controller chip. It has RAM slots for the systems Random Access Memory (SIMM or DIMM), and provides the systems chipset, controllers and underlying circuitry (bus system) to tie everything together.

The motherboard more or less is the computer. It defines the computer type upgradeability and expansion capability. Motherboards are of three basic types: non integrated motherboards, integrated motherboard and embedded motherboards.

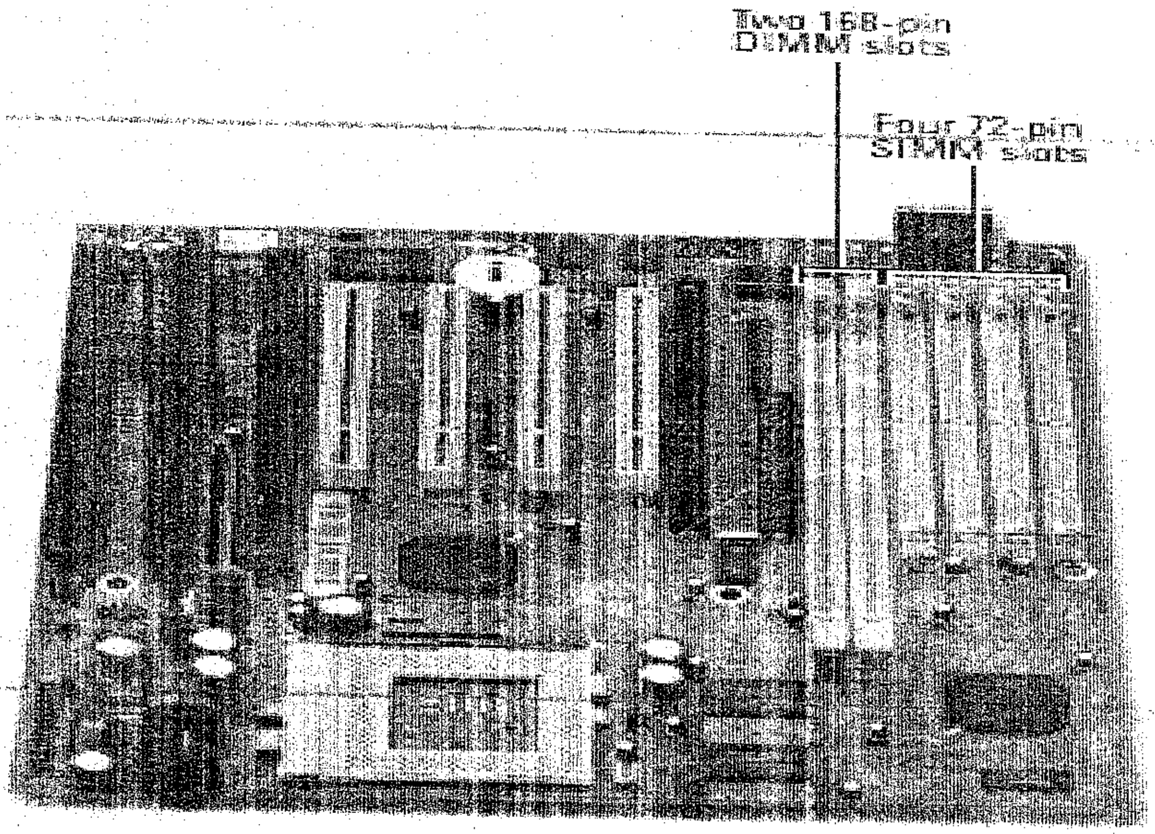


Fig 2.2: COMPUTER MOTHERBOARD

- **NON-INTEGRATED MOTHERBOARD:** These have assemblies such as the I/O port connectors (serial and parallel port), hard drive connectors or peddle boards, floppy controller and connectors, joystick connector e.t.c, installed as expansion boards. This takes up one or more of the motherboards expansion slots and reduces the amount of free space inside the computers case. Most of the older motherboards were non integrated.
- **INTEGRATED MOTHERBOARDS:** These have assemblies that are otherwise installed as expansion boards, integrated or built right onto the board. The serial and parallel ports, the IDE and floppy drive and joystick all connect directly to the motherboard. This is now standard on any late model 486 and above. It tends to free up some space inside the case and allows for better accessibility and air flow.

FORM FACTORS: Define the size, shape and screw placement on a motherboard since it is usually the technological advancements that have been achieved that allow for these changes, the form factors also define the technology to some extent. For now, form factors are categorized into three type; PC/XT, PC/baby AT and ATX.

- **PC/XT:** When IBM come out with its personal computer (PC), these were no standards and the motherboard tended to be a little on the large size with more space than it really needed. Within a short time , they had developed their extended technologies(XT) computer, reducing the size of the motherboard, to make it more compact and still accept the different circuits and components needed for the system. The XT quickly become a standard for motherboard in many of the clones that were being developed at the time.

- **AT/BABY AT:** Computers quickly become more and more powerful with features that required more, circuitry and components installed on the board. IBM had to increase the size of their boards to accept all these components and develop them.
- **ATX:** This is a form factor developed by Intel that closely conforms to the baby AT size. It put together some good ideas, engineering and design to make a standard that is cheaper to develop, allows for better component access and in some ways is faster and more stable. The ATX board measure approximately 9.5"x 12" and takes the baby AT board and turns it 90 degree to put the long edge of the board along the back of the computer case, which provides maximum space for expansion slots and I/O ports. The different I/O ports USB connector and PS/2 keyboard and mouse connector are stacked or "layered" and hardwire directly to the motherboard. The absence of a cable connector reduces radio interference as well as production costs. The ATX motherboard also defines the number and placement of mounting holes and uses a different power connection and different (PS/2) power supply.

OTHER MOTHERBOARD COMPONENTS: These include the small pin connectors that are used to connect and set the following controls and indicatory to the motherboard.

- Power Supply switch
- Reset switch
- The power on indicator
- Hard drive activity indicator
- In the case speaker connector

The motherboard manual needs to be consulted to see which connector are used for which item and how to hook them up. There is a bundle of cables near the front of the case (inside) which have labels on the connectors for these items.

2.3 THE CENTRAL PROCESSING UNIT (CPU)

The CPU is the main IC chip on the computers motherboard. They come in different shapes, sizes and package. Older CPUs came in DIP format, and somme 286s and early 386s where QSOPs, but they came as PGA or SPGA chips. The PGA chip used to be installed in a friction fit socket. Now they are installed on a zero insertion force (ZIP) socket.

The socket has a small handle or lever on the side of the socket that unclips and lift, releasing the pins completely. In both cases, pin No 1 is always designed both on the clip and on the socket. ~~There may be a dot in one corner, or an arrow or a small, silk screened~~ "1" usually, a CPU will have one of its corners leveled, and this has to line up with the designated corner on the socket. For Intels Pentium II, Pentium III and some of its Celeron chips, the CPU is put on a small circuit board with some external cache memory, and encased or packaged in a plastic cassette. This "slot 1" style CPU clips into a connector on the motherboard called a "slot 1" connector. Other mounting packages and some of the well known microprocessors that are mounted on them include:

- Socket 7 AMD K5, K6, Intel Pentium 75- 200Mhz, IBM
- Socket 370-some Intel Celerons
- Slot 1 – Intel Pentium II, Pentium III, some Celeron 266-533
- Slot II – intel Xeon
- Slot A- AMD Athlon

The socket 7 processors are becoming less popular. Os motherboards will accept more than one type of CPU but only accept one socket type. A specific CPU is described by its manufacturer, model and speed in mega hertz(Mhz), the system board has a quartz crystal clock whose cycle is measured in MHz (1,000,000Hz). A CPU runs at a multiple of the motherboard clock speed and the instructions it performs are synchronized with each cycle of the CPUs internal clock. Current technology by Intel and AMD provides us with up to 2.5GHz.

There are a lot of factors related to the CPU that have an effect on the speed and performance of computer system. Some of these factors include:

- Whether there is a math coprocessor present and if its internal or external
- The clock speed of the system and of the CPU
- The amount of internal cache and external cache available
- The bus architecture or supporting circuitry on the motherboard

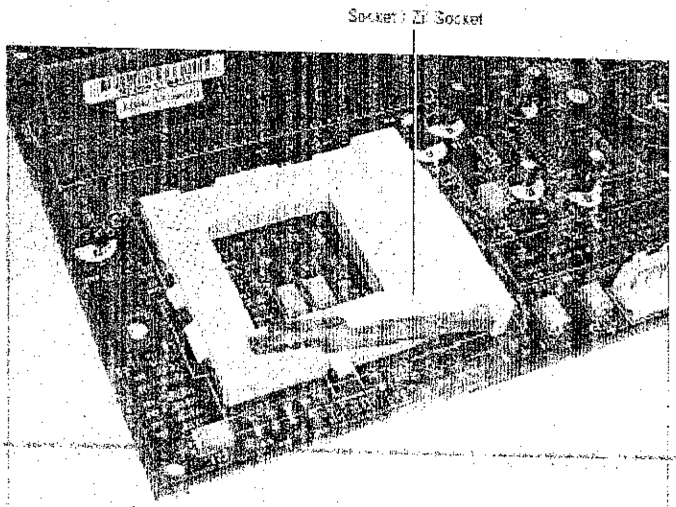


FIG 2.3: CENTRAL PROCESSING UNIT (CPU)

2.4 MEMORY

Computer memory is the mechanism by which the computer stores data for its use. The physical memory of the computer is classified to be either random access memory (RAM) or read only memory (ROM). A third category which is a special type of RAM known as *cache* memory, may also be classified as a type of memory.

RANDOM ACCESS MEMORY (RAM)

RAM is computer memory which can be read or changed by the user or computer. RAM chips are called DIPs which stands for *dual inline packages*. They are black chips with pins on both sides. To make memory installation easier than it was in the past, these DIP chips were place on modules. There are two main module types that memory comes packaged on today.

- Single inline memory module (SIMM): They may have DIPs on one or both sides and will have 30 or 72 pins. Today they normally are available in the 72 pin size which supports a 32bit data transfer between the microprocessor and the memory.
- Double inline memory module (DIMM): The modules have 168 pins and support a 64 bit data transfer between the microprocessor and the memory. Synchronous dynamic access memory (SDRAM) is the type of memory that is found on DIMM packages. The term SDRAM describes the memory type, and the term DIMM describes the package. These modules are available in 3.3 or 5Volt types and buffered or un buffered memory. This allows for choice of DIMM types. The choicest for today's motherboards is 3.3Volt unbuffered DIMMs.

To install these packages, press them into the socket on the motherboard and latch them in with a plastic latch on both sides. Normally as the memory module is pressed into place the latch will automatically latch the module in place. Here are some common types of dynamic Random Access Memory (DRAM).

- **FAST PAGE MODE DRAM (FPMDRAM):** When the first memory access is done. The row or page of the memory is specified. Once this is done, FPMDRAM allows any other row of memory to be accessed without specifying the row number. This speeds up access time.
- **EXTENDED DATA OUT DRAM (EDODRAM) :** This works like FPMDRAM but it holds the data valid even after signals have gone inactive. This allows the microprocessor to request memory, and it does not need to wait for the memory to become valid. I
- **SYNCHRONIZED DRAM (SDRAM):** This inputs and outputs its data synchronized to a clock that runs at some fraction of the microprocessor speed. SDRAM is faster than FPMDRAM and EDORAM. There is a new SDRAM called DDR (double data rate) SDRAM which allows data reads on both the rising and falling edge of the synchronize clock.
- **ROMBUS DRAM (RDRAM):** It uses a high bandwidth channel to transmit data at very high rates. It attempts to eliminate the time it takes to access memory.
- **SYNCLINK DRAM (SLDRAM):** It competes with RDRAM and uses 16 bank architecture rather than 4 along with other performance enhancing improvements.

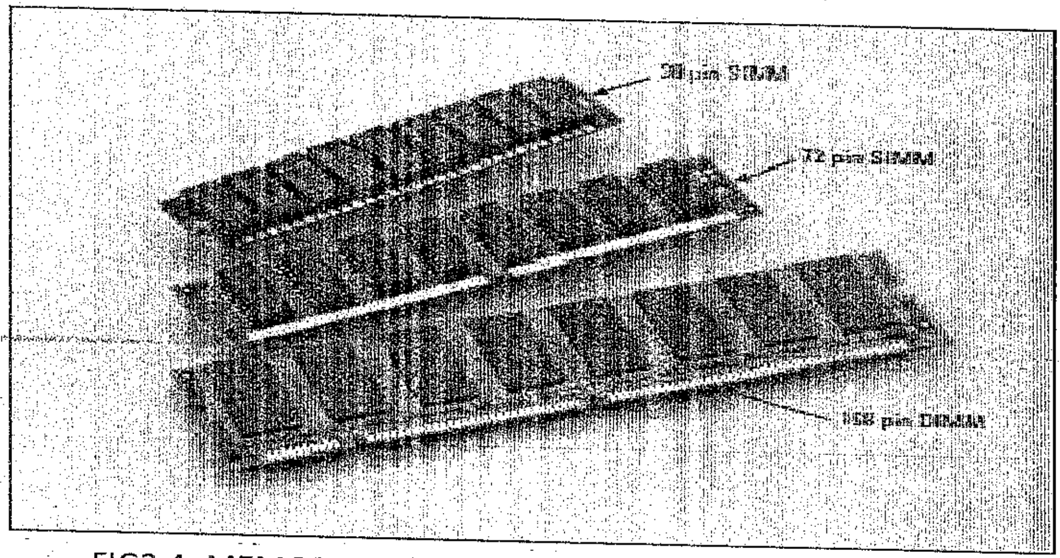


FIG2.4: MEMORY MODULES

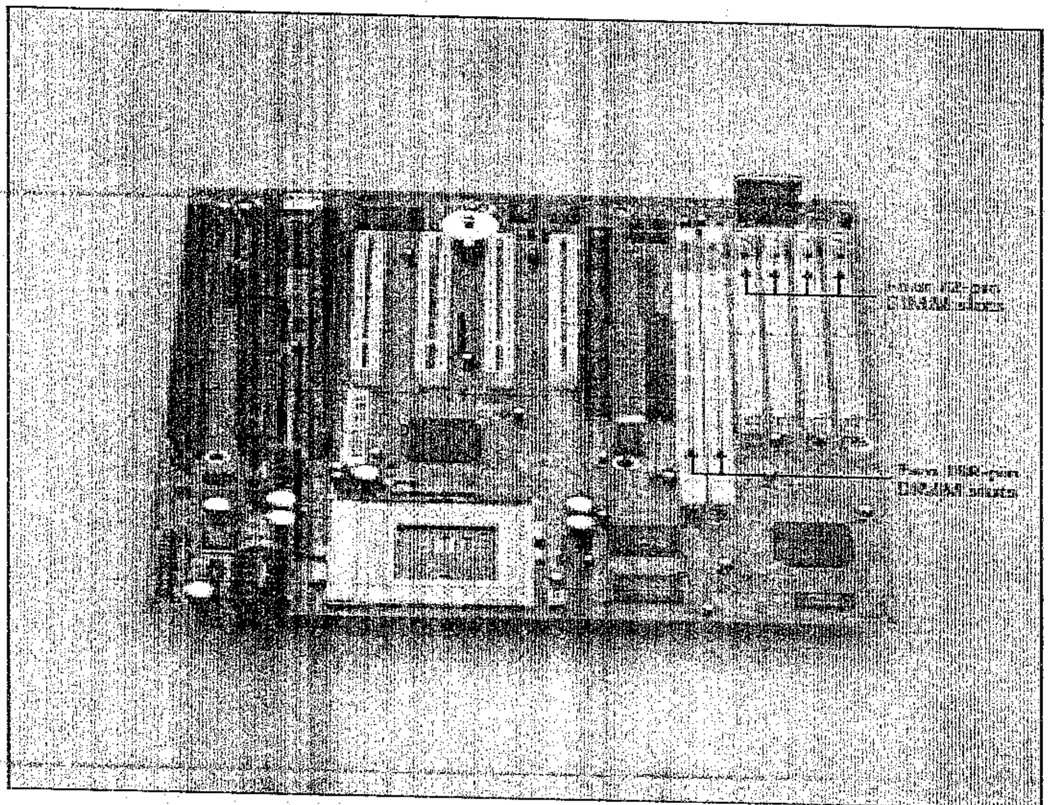


FIG 2.5: MEMORY SOCKETS

READ ONLY MEMORY (ROM)

ROM is computer memory which can be read by the computer but not altered in any way. The programming code and/or data on a ROM chip are written to the chip at the factory. It can be read, but it cannot be erased or removed. Its permanent. ROM chips are used to store permanent and critical information and programs that don't need to be changed, or don't need to be written to. Most personal computers have small amount of ROM that stores the code that starts up or boots the computer. Early computers also use ROM to store the BIOS which acts as a translator between the PCs hardware and the operating system. Here is some type of ROM:

- **PROM(PROGRAMMABLE READ ONLY MEMORY)**
- **EPROM (ERASABLE PROGRAMMABLE READ ONLY MEMORY)**
- **EEPROM (ELECTRICAL ERASABLE PROGRAMMABLE ROM)**
- **FLASH ROM (FLASH READ ONLY MEMORY)**

CACHE MEMORY

This is a special memory that operates much faster than SDRAM memory. It is not used for the entire system both for reason of expense and physical board and bus channel design requirements. Cache memory lies between the microprocessor and the system RAM. It is used as a buffer to reduce the microprocessor and level 2 (L2) which is just outside the microprocessor.

2.5 HARD DRIVE

The hard drive has five main component parts: platter, a spindle, read/write heads, head actuator and a circuit board.

TRACK: When a read/write head is stationary, the area that spins directly under the head is called a *track* there is the same number of tracks on both sides of each platter in a hard drive.
CYLINDER: The read/write heads are all positioned over the same track, on each side of each disk, simultaneously. The entire set of tracks moving under all the heads when they are stationary makes up the cylinder. Data is written from the outside cylinder inwards, using up space on each cylinder before the heads move to the next track, or next cylinder.

SECTOR: Each track is divided up into 512byte blocks called sectors. The data written to th hard drive is stored in these sectors, a cluster at a time.

CLUSTER: A defined number of sectors make up a cluster. The number of sector in each cluster varies depending upon the size of the hard drive and how it is partitioned.

STORAGE SPACE: Different hard drives have different numbers of sectors, tracks and platters. The total storage space on the hard drive is given by;

Storage space = sector size x sector per a tracks per a cylinder x the number of cylinder

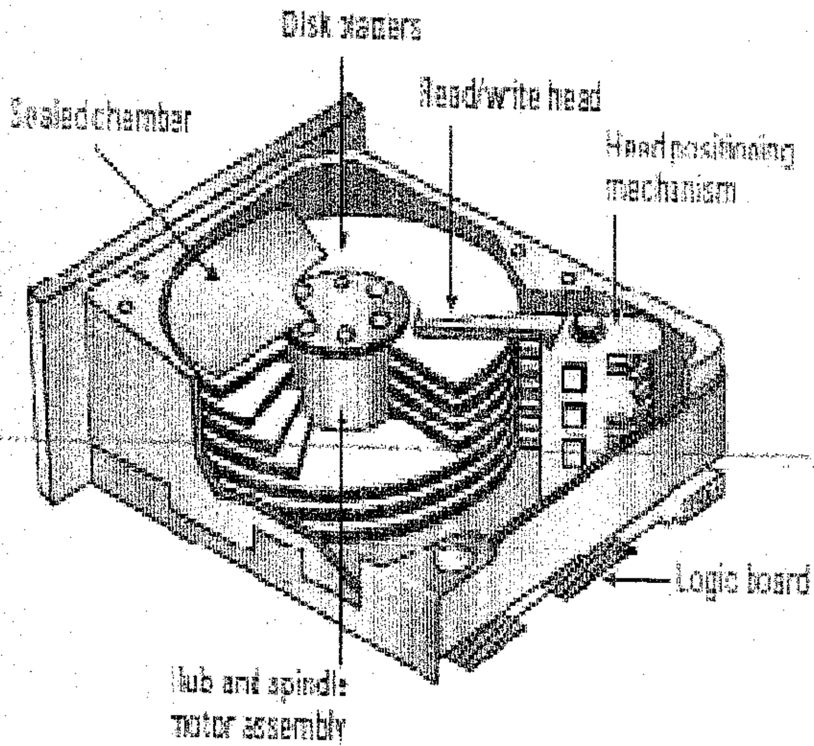


FIG 2.6: THE INTERNALS OF A HARD DRIVE

CONTROLER INTERFACE TYPES: A hard drive is a mass storage device where the operating system is installed along with many data files. There are two types of hard drives with regard to the controller.

INTEGRATED DRIVE TYPES: A controller based interface, if the primary concern for purchasing a computer is low price with reasonable performance, IDE is a good choice. It is still the most popular controller interface because of price.

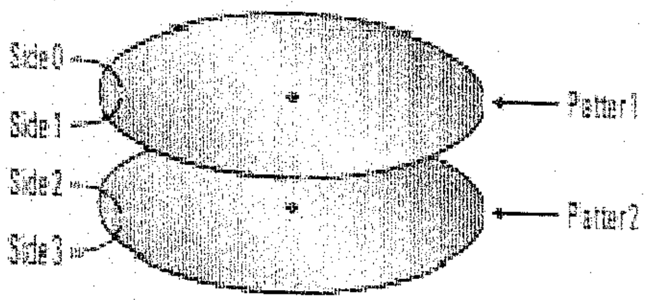
SMALL COMPUTER SYSTEM INTERFACE (SCSI): SCSI uses a separate bus hooked to the system bus using a host adapter. It is a more expensive system than IDE, but is better built and has a great deal of flexibility. They are mostly used by servers or high performance systems.

HARD DRIVE CHARACTERISTICS: Most hard drives have three basic charactistics or specification of main importance for performance. These are size, speed and reliability.

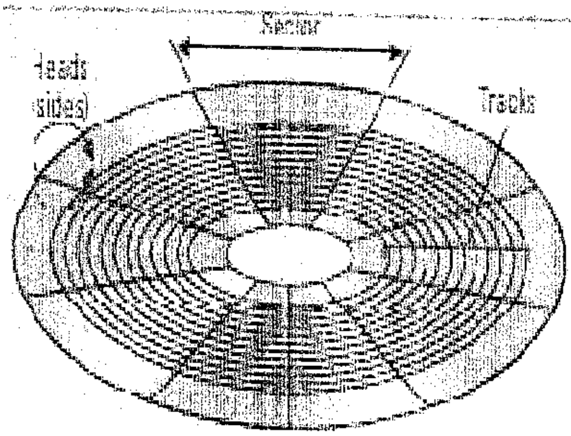
- **SIZE:** The size of the hard drive is expressed in terms of Giga bytes which is roughly 1000Megabyte. It is difficult to buy a drive less than 4 GB today. Typical sizes are 20 through 120 GB.
- **SPEED:** The data output of a hard drive is primary limited by the amount of time it takes for the electromagnetic head to reach the data at specific locations on the drive. The primary

factor of limitation is hard drive rotation speed. Common speeds today are 5400RPM (revolutions per minute), 7200RPM and 10000RPM

- **RELIABILITY:** The other performance factor that is worth considering is reliability. This is expressed as mean time between failures (MTBF). The higher the number, the better.



SIDES OF PLATTERS



TRACKS AND SECTOR ON DISK PLATTER

- **PLATTERS:** These are a series of aluminum alloy disks upon which data are stored. They are coated with a electromagnetic materials which can support magnetic states that are capable of being electrically altered. This means some type of electrical signal can alter the magnetic polarization of various area of the plates. The stat of these polarize area can also be sensed. Each platter can hold large amount of data.
- **SPINDLE:** Several platters are mounted on the spindle, which spins or rotates them. The spindle is run by a motor on the drive. It is kind of like the axle on the whel.
- **READ/WRITE HEADS:** The heads moves across the platter to write data onto the tracks of the platter.

- **HEAD ACTUATOR:** The hard drive head actuator control the read/write heads. The heads are at the end of an actuator arm which is attached to the actuator.
- **CIRCUIT BOARD:** This receives commands from the hard drive controller and translates them in order to move the head actuator which moves the read/write head across the platters to the required position. Each platter has data stored on it in a specific pattern for read and write access. The data is organized using low level formatting which is now done at the industry instead of individually as in the case with older drives.

2.6 CD ROM DRIVES

There are two primary type of CD ROMs today Read only CD ROM and Read and write CD ROM.

- **CD ROM SPEED:** The primary performance concern of CD ROM drives is their speed. Speeds are expressed in terms of 1x,2x,4x which is the number of times the drive is faster than the standard CD ROM reader (150,000 b/s). of the read only type, speeds have exceeded 50x. the read/write type of CDROM speeds is expressed with three values. They are read, write and rewrite. Current speeds of these devices are 48x, 32x, 10x. other types of storage devices include:
- **DVD DRIVE:** Most DVD drives also used the ATAPI interface. They are available as internal or external devices. They can operate at up to 16x speeds and are primarily used for video storage but the can be used to hold audio and computer data. DVD ROM device is for computer data storage. There are five recordable versions of DVD ROM. They all can read DVD ROM and DVD video discs, but different type of disc is used by each one for recording.
- **ZIP DRIVES:** A removable cartridge storage device that may be used to store compressed data as data backup. A zip drive between a 100Mb to 2 GB storage capacity. Some zip drives can also be used to read standard 3.5 inches floppy diskettes.

2.7 THE BUS

The bus is a network of wires or electronic pathway that lets the CPU sends various data values instructions, and information to all the devices and components inside the computer as well as the different peripherals and devices attached. A computers bus can be divided into two different types; *internal and external.*

INTERNAL BUS: The internal bus connects the different components inside the case, the CPU, system memory and all other components on the motherboard. Its also referred to as the system bus.

EXTERNAL BUS: The external bus connects the difficult external devices, peripherals, expansion slots, I/O ports and drive connections to the rest of the computer. The external bus, also called the expansion bus allows for the expansion of the computer capability. It is generally slower than the system bus.

EXPANSION SLOTS (CONNECTOR)

The expansion bus connects external devices (monitor, telephone line, printer, etc) to port on the back of the computer. There ports are actually parts of small circuit boards or cards that fit into connectors on the motherboard. The connectors are called expansion slots which are easy to recognize on the motherboard. They make up a row of long plastic connectors at the back of the computer with tiny copper 'finger slots' in a narrow channel that grab the metal fingers, or connectors on the expansion cards. The expansion card plug into the expansion slots which are attached to the expansion bus.

NOTE: Communication ports (com ports), printer port, hard drive and floppy connectors e.t.c, are all devices which used to be installed via adapter cards. These connectors are now integrated onto the motherboard, but they are still accessed via the expansion bus and are allocated the sam type of resources are required by expansion cards.

Technology has evolved in an effort to increase the speed, capacity and performance of expansion slots. This gave rise to various type of expansion buses, which are described by the type of connector and the particular technology or architecture used. Here are some of the common bus types.

8 BIT EXPANSION BUS: Early IBM PCs and XTs used the 8 bit expansion bus. That is, only eight data line run from the processor to the expansion connectors. It was a single long channel connector with 62 metal "finger connector" or channel. The bus speed was from 4.77 MHz to 8Mhz. it provided eight interrupts and four DMA channels which were all pre-assigned and was configured using jumpers and DIP switches.

16 BIT ISA BUS: The industry standard architecture (ISA) bus, also called the AT bus, had a 16 bit data path and provided sixteen interrupts and eight DMA channels. Many were pre assigned with a few open for expansion. It had a bus speed of 8 MHz and was capable of using 1 bus mastering device. The 16 bits ISA slot was characterized by 2 separate channel slots, a shorter one in front of the typical 8 bit slot to create the 16 bit connector. This allowed for backward compatibility with the older technology. The 16 bit ISA bus can be found in 286, 386, 486 and Pentium computers. IRQs and DMA channels were assign through jumpers on the the card itself and later through jumper jumper settings into an EEPROMM clip with some devices configured using software programs. ISA architecture also allowed for the faster CPU to have an internal clock or multiplier that was dissociated with the bus clock allowing each to run at its rated speed.

MICRO CHANNEL ARCHITECTURE (MCA) BUS: This was a 32 bit architecture that had a speed of 10MHz.The MCA bus was capable of using multiple mastering devices and could be configured using software.

VESA LOCAL BUS (VLB): The video electronics standards association (VESA) local bus had a 32 bit data path. It was building on the ISA architecture, but was longer, with a third connector that had all its lines running directly to the processor. It was usually used for video cards I/O cards and multimedia expansion cards. There couldn't be more than 3 VLB slots on a motherboard because the processor couldn't keep up with the transfer often; one of the slot was a shared slot.

PERIPHERAL COMPONENT INTERCONNECTS (PCI): The PCI bus was introduced with the Pentium computer. This bus supports both 32 and 64bit date path and uses a chipset that will also support ISA and EISA architectures. The PCI bus communicates with the processor through a bridge circuit, which acts like an interpreter. This means that it can be processor independent, PCI has a 33 MHz bus speed and can support multiple bus mastering devices. The cards are plug_and_play and come in two versions, 5Vdc and 3.3Vdc. The slots are keyed differently and will not allow the wrong voltage card to be inserted.

UNIVERSAL SERIEL BUS (USB): Universal serial bus is a relatively new bus technology. It was designed for low to mid speed peripherals such as scanners, keyboards, mice, joysticks, printers, modems and some CD ROMs. USB is plug_and_play and is completely "hot swappable".

PCMCIA BUS: PCMCIA stands for personal computer memory card international association. It is often called a PC card bus and is used in laptops and notebook computer to add external devices such as modems, network cards, memory and removable hard drives. The bus has a 16 bit data path and only supports one IRQ. It is software configurable; using what is called card and socket services software. There are three different types of PC cards and sockets.

- **TYPE I PC CARD:** This is 3.3mm thick and has a single row of connectors. It was used mainly for add on memory cards and isn't found in newer laptops.
- **TYPE II PC CARD:** This type is thicker (5mm) and has two rows of connectors. This is the most common and is used for modems and network interface cards. Most laptops today have two type II sockets.
- **TYPE III PC CARD:** These cards are not popular either they are thicker still (up to 10.5mm) and have three or four rows of connectors. They are used mainly for adding an external hard drive to a laptop or notebook computer.

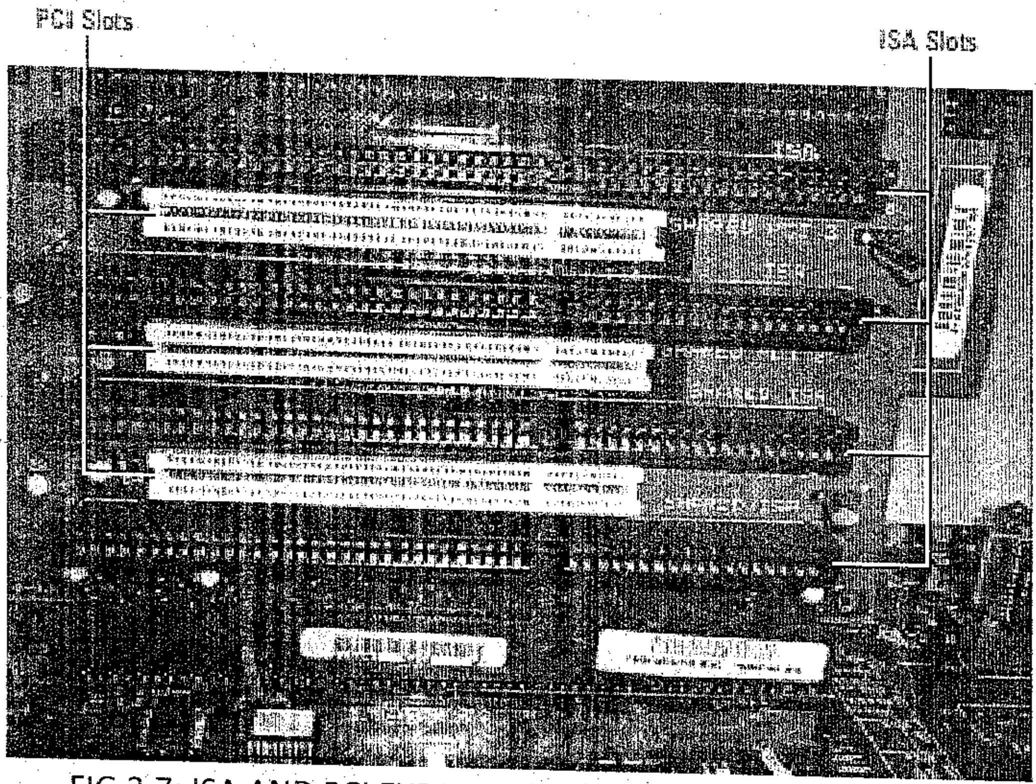


FIG 2.7: ISA AND PCI EXPANSION SLOTS

2.8 MONITOR

The computer monitor is an output device that is part of the computer display system. A cable connects it to a video adapter (video card) this is installed in an expansion slot on the motherboard. This system converts signals into text and pictures and display them on a TV- like screen (the monitor).

CATHOD RAY TUBE (CRT): The CRT is the "picture tube" of the monitor. It is a large vacuum tube that is shaped like a bottle. The tube tapes near the back of the positive charged cathode, or "electron gun". The electron gun shoots electrons at the back of the positive charged screen, which is coated with phosphorus chemical. This excites the phosphorus causing them to glow as individual dots called *pixels* (picture elements). The image seen on the monitor screen are made of thousand of tiny dots (pixels) the distance between the pixel determine the quality of the image, the closer the pixels are, the sharper the image on screen. The distance between pixels is called "dot pitch" and is measured in millimeter. There are a couple of electromagnets (yokes) around the collar of the tube that actually bend the beam of electrons. The beam scans across the monitor from left to right and top to bottom to create, or draw the image, line by line. If the scanning beam hits each and every line of pixels in succession, on each pass, then the monitor is known as a non interlaced monitor. The electron beam on an interlaced monitor scans the odd numbered lines on